

10Pearls DataScience Project

AQI PREDICTOR

Real-Time Air Quality Forecasting for Karachi

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Project Overview

01 Objective

Predict AQI for the next 72 hours using a fully serverless ML pipeline.

02 Stack

Python, Scikit-learn, Ridge Regression, Hopsworks, Streamlit, Docker, GitHub Actions.

03 Features

Automated data ingestion, feature engineering, model training, and interactive visualization.

Data & EDA

Data Source

Open-Meteo API for AQI and weather data (PM2.5, PM10, CO, NO₂, O₃, SO₂, NH₃, temperature, humidity, wind speed, precipitation).

EDA Insights

AQI spikes correlate with low wind speed and high PM2.5.

Hour-of-day and day-of-week strongly affect AQI.

Temperature and humidity show moderate correlation with AQI.

Feature Engineering

Time-based features hour, day, month, seasonal encoding

Derived features AQI change rate, lag values, 3-hour & 24-hour rolling means

Weather-pollution interactions wind × pollutant, temperature × humidity

Additional Feature importance ranked via SHAP for explainability
All features stored in Hopsworks Feature Store

Model Training

Models tested

Random Forest, Ridge Regression, Gradient Boosting

Evaluation metrics

RMSE, MAE, R^2

Best-performing model

Ridge Regression for short-term AQI forecasting

Additional

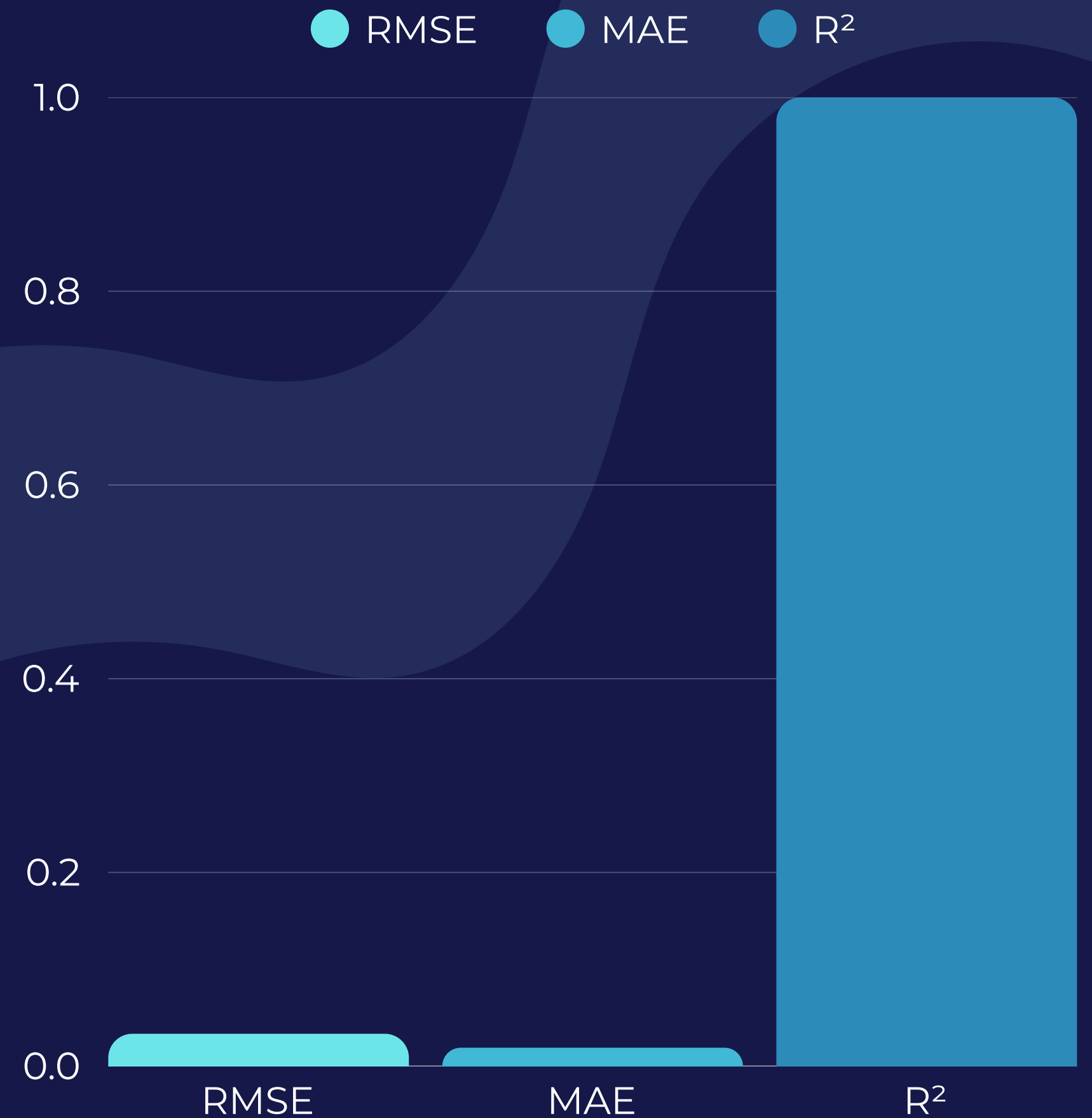
training pipeline automated via GitHub Actions; model saved to Hopsworks Model Registry

Model Evaluation

Feature importance
(via SHAP)

Top predictors: PM2.5, AQI change
rate, hour-of-day

RMSE, MAE, R^2 metrics for 72-
hour forecast



Dashboard & Deployment

Streamlit dashboard

features:

- Live AQI and pollutant breakdown
- 72-hour forecast visualization
- SHAP-based feature importance plots

Docker

Docker containerization ensures scalability and reproducibility

CI/CD: GitHub Actions

- Hourly feature updates
- Daily model retraining

Limitations

Model is Karachi-specific; other cities need retraining

Sudden pollution spikes may reduce forecast accuracy

Dependent on API data availability

Deep learning models optional, not deployed

Extras / Future Work

Alerts for hazardous AQI levels

Multi-city support

LSTM / Transformer for long-term
forecasts

Expanded SHAP/LIME
explainability

Conclusion

Fully automated, scalable, serverless
AQI prediction system

Real-time insights for
environmental awareness and
public safety

Combines ML, cloud
automation, and interactive
dashboards for decision-making

